- Many insulators are not colored because their band gaps are so large that no visible light is absorbed. Adding dopants can introduce donor or acceptor states that enable absorption of visible light, so the doped materials are colored.
- Phosphors are wide band gap materials doped to emit light at specific wavelengths upon excitation by electron impact or incident light at other wavelengths. Phosphors are used to generate visible light in fluorescent fixtures, and to display signals on video monitors.

PROBLEMS

Answers to problems whose numbers are boldface appear in Appendix G. Problems that are more challenging are indicated with asterisks.

Minerals: Naturally Occurring Inorganic Materials

1. Draw a Lewis electron-dot diagram for the disilicate ion ($Si_2O_7^{6-}$). What changes in this structure would be necessary to produce the structure of the pyrophosphate ion ($P_2O_7^{4-}$) and the pyrosulfate ion ($S_2O_7^{2-}$)? What is the analogous compound of chlorine?

2. Draw a Lewis electron-dot diagram for the cyclosilicate ion ($Si_6O_{18}^{12-}$), which forms part of the structures of beryl and emerald.

3. Using Table 22.1, predict the structure of each of the following silicate minerals (network, sheets, double chains, and so forth). Give the oxidation state of each atom.
(a) Andradite, $Ca_3Fe_2(SiO_4)_3$
(b) Vlasovite, $Na_2ZrSi_4O_{10}$
(c) Hardystonite, $Ca_2ZnSi_2O_7$
(d) Chrysotile, $Mg_3Si_2O_5(OH)_4$

4. Using Table 22.1, predict the structure of each of the following silicate minerals (network, sheets, double chains, and so forth). Give the oxidation state of each atom.
(a) Tremolite, $Ca_2Mg_5(Si_4O_{11})_2(OH)_2$
(b) Gillespite, $BaFeSi_4O_{10}$
(c) Uvarovite, $Ca_3Cr_2(SiO_4)_3$
(d) Barysilate, $MnPb_8(Si_2O_7)_3$

5. Using Table 22.1, predict the structure of each of the following aluminosilicate minerals (network, sheets, double chains, and so forth). In each case, the Al atoms grouped with the Si and O in the formula substitute for Si in tetrahedral sites. Give the oxidation state of each atom.
(a) Keatite, $Li(AlSi_2O_6)$
(b) Muscovite, $KAl_2(AlSi_3O_{10})(OH)_2$
(c) Cordierite, $Al_3Mg_2(AlSi_5O_{18})$

6. Using Table 22.1, predict the structure of each of the following aluminosilicate minerals (network, sheets, double chains, and so forth). In each case, the Al atoms grouped with the Si and O in the formula substitute for Si in tetrahedral sites. Give the oxidation state of each atom.
(a) Amesite, $Mg_2Al(AlSiO_5)(OH)_4$
(b) Phlogopite, $KMg_3(AlSi_3O_{10})(OH)_2$
(c) Thomsonite, $NaCa_2(Al_5Si_5O_{20}) \cdot 6\ H_2O$

Silicate Ceramics

7. A ceramic that has been much used by artisans and craftsmen for the carving of small figurines is based on the mineral steatite (commonly known as soapstone). Steatite is a hydrated magnesium silicate that has the composition $Mg_3Si_4O_{10}(OH)_2$. It is remarkably soft—a fingernail can scratch it. When heated in a furnace to about 1000°C, chemical reaction transforms it into a hard, two-phase composite of magnesium silicate ($MgSiO_3$) and quartz in much the same way that clay minerals are converted into mullite ($Al_6Si_2O_{13}$) and cristobalite (SiO_2) on firing. Write a balanced chemical equation for this reaction.

8. A clay mineral that is frequently used together with or in place of kaolinite is pyrophyllite ($Al_2Si_4O_{10}(OH)_2$). Write a balanced chemical equation for the production of mullite and cristobalite on the firing of pyrophyllite.

9. Calculate the volume of carbon dioxide produced at standard temperature and pressure when a sheet of ordinary glass of mass 2.50 kg is made from its starting materials—sodium carbonate, calcium carbonate, and silica. Take the composition of the glass to be $Na_2O{\cdot}CaO{\cdot}(SiO_2)_6$.

10. Calculate the volume of steam produced when a 4.0-kg brick made from pure kaolinite is completely dehydrated at 600°C and a pressure of 1.00 atm.

11. A sample of soda-lime glass for tableware is analyzed and found to contain the following percentages by mass of oxides: SiO_2, 72.4%; Na_2O, 18.1%; CaO, 8.1%; Al_2O_3, 1.0%; MgO, 0.2%; BaO, 0.2%. (The elements are not actually present as binary oxides, but this is the way compositions are usually given.) Calculate the chemical amounts of Si, Na, Ca, Al, Mg, and Ba atoms per mole of O atoms in this sample.

12. A sample of Portland cement is analyzed and found to contain the following percentages by mass of oxides: CaO, 64.3%; SiO_2, 21.2%; Al_2O_3, 5.9%; Fe_2O_3, 2.9%; MgO, 2.5%; SO_3, 1.8%; Na_2O, 1.4%. Calculate the chemical amounts of Ca, Si, Al, Fe, Mg, S, and Na atoms per mole of O atoms in this sample.

13. The most important contributor to the strength of hardened Portland cement is tricalcium silicate, $(CaO)_3{\cdot}SiO_2$, for which the measured standard enthalpy of formation is -2929.2 kJ mol^{-1}. Calculate the standard enthalpy change for the production of 1.00 mol tricalcium silicate from quartz and lime.

14. One of the simplest of the heat-generating reactions that occur when water is added to cement is the production of calcium hydroxide (slaked lime) from lime. Write a balanced chemical equation for this reaction, and use data from Appendix D to calculate the amount of heat generated by the reaction of 1.00 kg lime with water at room conditions.

Copyright 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. Due to electronic rights, some third party content may be suppressed from the eBook and/or eChapter(s).
Editorial review has deemed that any suppressed content does not materially affect the overall learning experience. Cengage Learning reserves the right to remove additional content at any time if subsequent rights restrictions require it.

Nonsilicate Ceramics

15. Calculate the average oxidation number of the copper in $YBa_2Cu_3O_{9-x}$ if $x = 2$. Assume that the rare-earth element yttrium is in its usual +3 oxidation state.

16. The mixed oxide ceramic $Tl_2Ca_2Ba_2Cu_3O_{10+x}$ has zero electrical resistance at 125 K. Calculate the average oxidation number of the copper in this compound if $x = 0.50$ and thallium is in the +3 oxidation state.

17. Silicon carbide (SiC) is made by the high-temperature reaction of silica sand (quartz) with coke; the byproduct is carbon monoxide.
(a) Write a balanced chemical equation for this reaction.
(b) Calculate the standard enthalpy change per mole of SiC produced.
(c) Predict (qualitatively) the following physical properties of silicon carbide: conductivity, melting point, and hardness.

18. Boron nitride (BN) is made by the reaction of boron trichloride with ammonia.
(a) Write a balanced chemical equation for this reaction.
(b) Calculate the standard enthalpy change per mole of BN produced, given that the standard molar enthalpy of formation of BN(*s*) is (254.4 kJ mol^{-1}.
(c) Predict (qualitatively) the following physical properties of boron nitride: conductivity, melting point, and hardness.

19. The standard free energy of formation of cubic silicon carbide (SiC) is 62.8 kJ mol^{-1}. Determine the standard free energy change when 1.00 mol SiC reacts with oxygen to form SiO_2 (*s*, quartz) and $CO_2(g)$. Is silicon carbide thermodynamically stable in the air at room conditions?

20. The standard free energy of formation of boron carbide (B_4C) is (71 kJ mol^{-1}. Determine the standard free energy change when 1.00 mol B_4C reacts with oxygen to form $B_2O_3(s)$ and $CO_2(g)$. Is boron carbide thermodynamically stable in the air at room conditions?

Electrical Conduction in Materials

21. A cylindrical sample of solid germanium has length 55.0 mm and diameter 5.0 mm. In a test circuit, 0.150 A of current flowed through this sample when the voltage applied between its ends was 17.5 V. What is the electrical conductivity of this sample?

22. A gold wire 4.0 mm in diameter and 1.5 m in length is to be used in a test circuit. (a) Calculate the resistance of the wire. (b) Calculate the current density in the wire when the voltage applied between its ends is 0.070 V. (c) Calculate the electric field in the wire.

23. The mobilities for Na^+ and Cl^- in aqueous solution are given in Table 22.7. Calculate the conductivity of a 0.10-M solution of NaCl in water at 25°C.

24. Explain why the ionic mobility of CH_3COO^- is smaller than that for Cl^-.

25. The electrical conductivity for copper is given in Table 22.6. The electron mobility in copper at room temperature is 3.0×10^{-3} m^2 V^{-1} s^{-1}. Using the Drude model for metallic conductivity, calculate the number of free electrons per Cu atom. The density of copper is 8.9 g cm^{-3}.

26. A variety of useful metallic alloys can be prepared by dissolving Ni in Cu. The room temperature resistivity of pure copper is 1.6×10^{-8} Ω m. As nickel is dissolved in copper up to 50% mass, the resistivity increases in a nearly linear fashion to the value 47.0×10^{-8} Ω m. Explain this increase qualitatively.

Band Theory of Conduction

27. Electrons in a semiconductor can be excited from the valence band to the conduction band through the absorption of photons with energies exceeding the band gap. At room temperature, indium phosphide (InP) is a semiconductor that absorbs light only at wavelengths less than 920 nm. Calculate the band gap in InP.

28. Both GaAs and CdS are semiconductors that are being studied for possible use in solar cells to generate electric current from sunlight. Their band gaps are 2.29×10^{-19} J and 3.88×10^{-19} J, respectively, at room temperature. Calculate the longest wavelength of light that is capable of exciting electrons across the band gap in each of these substances. In which region of the electromagnetic spectrum do these wavelengths fall? Use this result to explain why CdS-based sensors are used in some cameras to estimate the proper exposure conditions.

29. The number of electrons excited to the conduction band per cubic centimeter in a semiconductor can be estimated from the following equation:

$$n_e = (4.8 \times 10^{15}\ \text{cm}^{-3}\ \text{K}^{-3/2})\, T^{3/2}\, e^{-E_g/(2RT)}$$

where T is the temperature in kelvins and E_g the band gap in joules *per mole*. The band gap of diamond at 300 K is 8.7×10^{-19} J. How many electrons are thermally excited to the conduction band at this temperature in a 1.00-cm^3 diamond crystal?

30. The band gap of pure crystalline germanium is 1.1×10^{-19} J at 300 K. How many electrons are excited from the valence band to the conduction band in a 1.00-cm^3 crystal of germanium at 300 K? Use the equation given in the preceding problem.

Semiconductors

31. Describe the nature of electrical conduction in (a) silicon doped with phosphorus and (b) indium antimonide doped with zinc.

32. Describe the nature of electrical conduction in (a) germanium doped with indium and (b) cadmium sulfide doped with arsenic.

33. In a light-emitting diode (LED), which is used in displays on electronic equipment, watches, and clocks, a voltage is imposed across an *n–p* semiconductor junction. The electrons on the *n* side combine with the holes on the *p* side and emit light at the frequency of the band gap. This process can also be described as the emission of light as electrons fall from levels in the conduction band to empty levels in the valence band. It is the reverse of the production of electric current by illumination of a semiconductor.

Many LEDs are made from semiconductors that have the general composition $GaAs_{1-x}P_x$. When x is varied between 0 and 1, the band gap changes and, with it, the color of light emitted by the diode. When $x = 0.4$, the band gap is 2.9×10^{-19} J. Determine the wavelength and color of the light emitted by this LED.

Copyright 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. Due to electronic rights, some third party content may be suppressed from the eBook and/or eChapter(s).
Editorial review has deemed that any suppressed content does not materially affect the overall learning experience. Cengage Learning reserves the right to remove additional content at any time if subsequent rights restrictions require it.

34. When the LED described in Problem 33 has the composition $GaAs_{0.14}P_{0.86}$ (i.e., $x = 0.86$), the band gap has increased to 3.4×10^{-19} J. Determine the wavelength and color of the light emitted by this LED.

Pigments and Phosphors: Optical Displays

35. The pigment zinc white (ZnO) turns bright yellow when heated, but the white color returns when the sample is cooled. Does the band gap increase or decrease when the sample is heated?

36. Mercury(II) sulfide (HgS) exists in two different crystalline forms. In cinnabar, the band gap is 3.2×10^{-19} J; in metacinnabar, it is 2.6×10^{-19} J. In some old paintings with improperly formulated paints, the pigment vermilion (cinnabar) has transformed to metacinnabar on exposure to light. Describe the color change that results.

ADDITIONAL PROBLEMS

37. Predict the structure of each of the following silicate minerals (network, sheets, double chains, and so forth). Give the oxidation state of each atom.
    (a) Apophyllite, $KCa_4(Si_8O_{20})F \cdot 8\ H_2O$
    (b) Rhodonite, $CaMn_4(Si_5O_{15})$
    (c) Margarite, $CaAl_2(Al_2Si_2O_{10})(OH)_2$

38. Using Table 22.1, predict the kind of structure formed by manganpyrosmalite, a silicate mineral with chemical formula $Mn_{12}FeMg_3(Si_{12}O_{30})(OH)_{10}Cl_{10}$. Give the oxidation state of each atom in this formula unit.

39. A reference book lists the chemical formula of one form of vermiculite as

$$[(Mg_{2.36}Fe_{0.48}Al_{0.16})(Si_{2.72}Al_{1.28})O_{10}(OH)_2][Mg_{0.32}(H_2O)_{4.32}]$$

Determine the oxidation state of the iron in this mineral.

40. The most common feldspars are those that contain potassium, sodium, and calcium cations. They are called, respectively, orthoclase ($KAlSi_3O_8$), albite ($NaAlSi_3O_8$), and anorthite ($CaAl_2Si_2O_8$). The solid solubility of orthoclase in albite is limited, and its solubility in anorthite is almost negligible. Albite and anorthite, however, are completely miscible at high temperatures and show complete solid solution. Offer an explanation for these observations, based on the tabulated radii of the K^+, Na^+, and Ca^{2+} ions from Appendix F.

41. The clay mineral kaolinite ($Al_2Si_2O_5(OH)_4$) is formed by the weathering action of water containing dissolved carbon dioxide on the feldspar mineral anorthite ($CaAl_2Si_2O_8$). Write a balanced chemical equation for the reaction that occurs. The CO_2 forms H_2CO_3 as it dissolves. As the pH is lowered, will the weathering occur to a greater or a lesser extent?

42. Certain kinds of zeolite have the general formula $M_2O{\cdot}Al_2O_3{\cdot}ySiO_2{\cdot}wH_2O$, where M is an alkali metal such as sodium or potassium, y is 2 or more, and w is any integer. Compute the mass percentage of aluminum in a zeolite that has M = K, $y = 4$, and $w = 6$.

43. (a) Use data from Tables 15.2 and 16.2 to calculate the solubility of $CaCO_3$ in water at pH 7.
    (b) Will the solubility increase or decrease if the pH is lowered and the water becomes more acidic?
    (c) Calculate the maximum amount of limestone (primarily calcium carbonate) that could dissolve per year in a river at pH 7 with an average flow rate of 1.0×10^6 m^3/h.

44. Silica (SiO_2) exists in several forms, including quartz (molar volume 22.69 $cm^3\ mol^{-1}$) and cristobalite (molar volume 25.74 $cm^3\ mol^{-1}$).
    (a) Use data from Appendix D to calculate $\Delta H°$, $\Delta S°$, and ΔG at 25°C.
    (b) Which form is thermodynamically stable at 25°C?
    (c) Which form is stable at very high temperatures, provided that melting does not occur first?

45. Talc, $Mg_3Si_4O_{10}(OH)_2$, reacts with forsterite (Mg_2SiO_4) to form enstatite ($MgSiO_3$) and water vapor.
    (a) Write a balanced chemical equation for this reaction.
    (b) If the water pressure is equal to the total pressure, will formation of products be favored or disfavored with increasing total pressure?
    (c) The entropy change for this reaction is positive. Will the slope of the coexistence curve (pressure plotted against temperature) be positive or negative?

46. In what ways does soda-lime glass resemble and in what ways does it differ from a pot made from the firing of kaolinite? Include the following aspects in your discussion: composition, structure, physical properties, and method of preparation.

47. Iron oxides are red when the average oxidation state of iron is high and black when it is low. To impart each of these colors to a pot made from clay that contains iron oxides, would you use an air-rich or a smoky atmosphere in the kiln? Explain.

48. Refractories can be classified as acidic or basic, depending on the properties of the oxides in question. A basic refractory must not be used in contact with acid, and an acidic refractory must not be used in contact with a base. Classify magnesia and silica as acidic or basic refractories.

49. Dolomite bricks are used in the linings of furnaces in the cement and steel industries. Pure dolomite contains 45.7% $MgCO_3$ and 54.3% $CaCO_3$ by mass. Determine the empirical formula of dolomite.

50. Beryllia (BeO) ceramics have some use but show only poor resistance to strong acids and bases. Write likely chemical equations for the reaction of BeO with a strong acid and with a strong base.

51. Silicon nitride resists all acids except hydrofluoric, with which it reacts to give silicon tetrafluoride and ammonia. Write a balanced chemical equation for this reaction.

52. Compare oxide ceramics such as alumina (Al_2O_3) and magnesia (MgO), which have significant ionic character, with covalently bonded nonoxide ceramics such as silicon car-

Copyright 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. Due to electronic rights, some third party content may be suppressed from the eBook and/or eChapter(s).
Editorial review has deemed that any suppressed content does not materially affect the overall learning experience. Cengage Learning reserves the right to remove additional content at any time if subsequent rights restrictions require it.

bide (SiC) and boron carbide (B_4C; see Problems 19 and 20) with respect to thermodynamic stability at ordinary conditions.

53. Compare the hybridization of Si atoms in Si(*s*) with that of C atoms in graphite (see Fig. 21.26). If silicon were to adopt the graphite structure, would its electrical conductivity be high or low?

54. Describe how the band gap varies from a metal to a semiconductor to an insulator.

55. Suppose some people are sitting in a row at a movie theater, with a single empty seat on the left end of the row. Every 5 minutes, a person moves into a seat on his or her left if it is empty. In what direction and with what speed does the empty seat "move" along the row? Comment on the connection with hole motion in *p*-type semiconductors.

56. A sample of silicon doped with antimony is an *n*-type semiconductor. Suppose a small amount of gallium is added to such a semiconductor. Describe how the conduction properties of the solid will vary with the amount of gallium added.

Copyright 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. Due to electronic rights, some third party content may be suppressed from the eBook and/or eChapter(s).
Editorial review has deemed that any suppressed content does not materially affect the overall learning experience. Cengage Learning reserves the right to remove additional content at any time if subsequent rights restrictions require it.